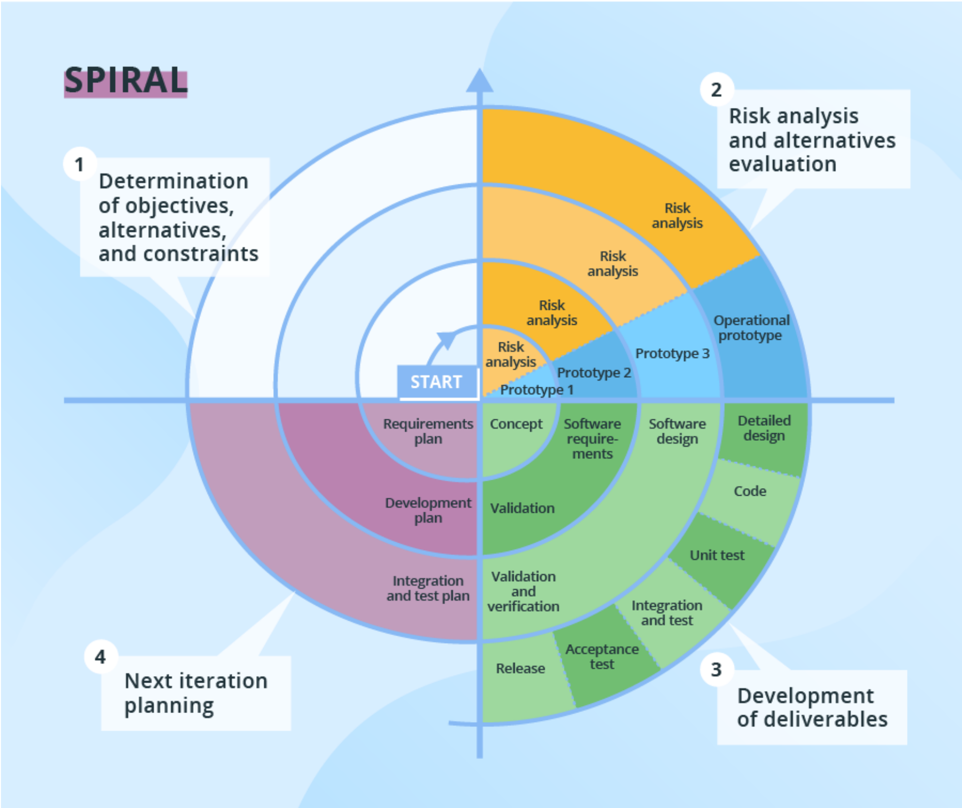
Spiral Model

Risk-Focused Development



The **Spiral Model** emphasizes risk analysis and iterative refinement through continuous cycles. Each cycle involves four major phases — **planning**, **risk analysis**, **prototype development**, and **evaluation**. With every iteration, the system evolves, incorporating feedback and minimizing project risks.

Key Characteristics

- Each spiral cycle results in a more complete and refined version of the software.
- Combines elements of both iterative and prototyping approaches.
- Focuses heavily on identifying and managing project risks early.
- Encourages active client involvement throughout the process.

Benefits

- Provides strong risk control through early identification and mitigation.
- Enables flexibility to adjust requirements across cycles.
- Enhances reliability and quality via repeated refinements.

Challenges

- Requires significant resources and expertise in risk management.
- Demands consistent client involvement at every stage.
- Difficult to estimate total project cost and timeline early on.
- Changes identified mid-cycle are deferred to the next iteration.

Best Suited For

- Projects with unclear or evolving business goals.
- High-risk or complex projects such as ERP systems or banking software.
- Research and Development (R&D) or experimental software initiatives.

Not Ideal For

- Small-scale or low-risk projects with fixed requirements.
- Projects with limited budgets or short timelines.

Summary

The Spiral Model combines iterative development with risk-driven planning and evaluation. It ensures reliability and flexibility but demands high cost, continuous stakeholder engagement, and mature project management practices.